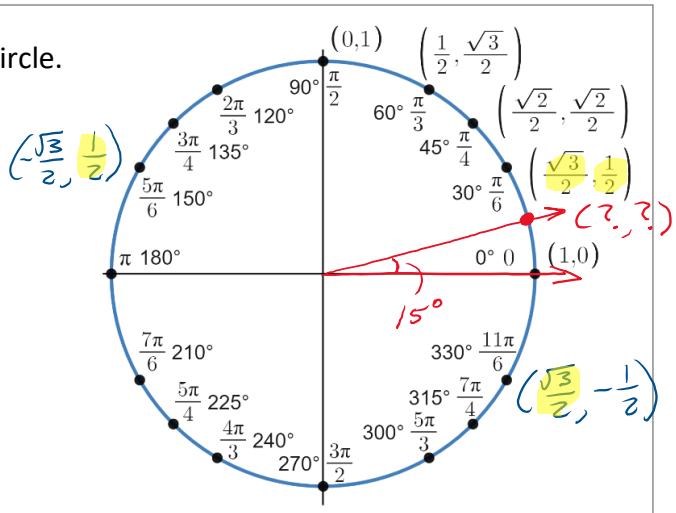
Sum and Difference Identities

Motivating Example

Before we dive into the sum and difference identities, let's be reminded of the process for evaluating trigonometric functions using the unit circle.

EXAMPLE 1

Evaluate each expression using the unit circle.



(a) $\sin\left(\frac{17\pi}{6}\right)$ $= \sin\left(\frac{5\pi}{6}\right) = \frac{1}{2}$

$\frac{17\pi}{6} - \frac{12\pi}{6} = \frac{5\pi}{6}$

(b) $\cos(1050°)$ $= \cos(330°) = \frac{\sqrt{3}}{2}$

$1050 - 360 = 690$

$690 - 360 = 330$

(c) $\tan\left(-\frac{11\pi}{6}\right)$ $= \tan\left(\frac{\pi}{6}\right) = \frac{\sin(\pi/6)}{\cos(\pi/6)} = \frac{1/2}{\sqrt{3}/2} = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$

$-\frac{11\pi}{6} + \frac{12\pi}{6} = \frac{\pi}{6}$

(d) $\sin(15°) = ?$ The standard unit circle does not yield a value for $\theta = 15°$

$0 < \sin(15°) < \frac{1}{2}$

In the last example above, we are not able, at least not with our current methods, to evaluate the expression exactly. However, when this happens, it may be possible to apply identities to rewrite the expression into an equivalent form where the unit circle can again be helpful.

The Sum and Difference Identities

PROPERTY: ***The Sum and Difference Identities***	
Let θ be any value for which the following functions are defined.	
Sine of a Sum or Difference	$\sin(A+B) = \sin(A)\cos(B) + \cos(A)\sin(B)$ $\sin(A-B) = \sin(A)\cos(B) - \cos(A)\sin(B)$
Cosine of a Sum or Difference	$\cos(A+B) = \cos(A)\cos(B) - \sin(A)\sin(B)$ $\cos(A-B) = \cos(A)\cos(B) + \sin(A)\sin(B)$
Tangent of a Sum or Difference	$\tan(A+B) = \dfrac{\tan(A)+\tan(B)}{1-\tan(A)\tan(B)}$ $\tan(A-B) = \dfrac{\tan(A)-\tan(B)}{1+\tan(A)\tan(B)}$

EXAMPLE 2

Use the sum identities, along with the even/odd identities, to verify the difference identities.

$\sin(A-B) =$ $\sin(A+{}^{-}B) = \sin(A)\cos(-B) + \cos(A)\sin(-B)$

$= \sin(A)\cos(B) - \cos(A)\sin(B)$

$\cos(A-B) =$ $\cos(A+{}^{-}B) = \cos(A)\cos(-B) - \sin(A)\sin(-B)$

$= \cos(A)\cos(B) - {}^{-}\sin(A)\sin(B)$

$= \cos(A)\cos(B) + \sin(A)\sin(B)$

$\tan(A-B) =$ $\tan(A+{}^{-}B) = \dfrac{\tan(A)+\tan(-B)}{1-\tan(A)\tan(-B)}$

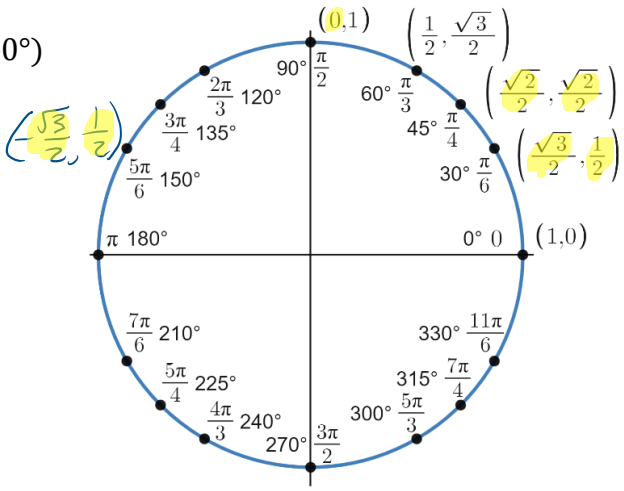
$= \dfrac{\tan(A)-\tan(B)}{1-{}^{-}\tan(A)\tan(B)} = \dfrac{\tan(A)-\tan(B)}{1+\tan(A)\tan(B)}$

Evaluating Trigonometric Expressions

These new identities extend our ability to evaluate trigonometric functions to a greater variety of angle measures. For large positive or negative angles, it may be helpful to add/subtract by one or more full rotations to find coterminal angles that may be more convenient to work with.

EXAMPLE 3

Evaluate each expression by applying the sum/difference identities and the unit circle.



(a) $\sin(110°)\cos(40°) + \cos(110°)\sin(40°)$

$= \sin(110° + 40°)$

$= \sin(150°) = \frac{1}{2}$

(b) $\cos\left(\frac{\pi}{12}\right)\cos\left(\frac{5\pi}{12}\right) - \sin\left(\frac{\pi}{12}\right)\sin\left(\frac{5\pi}{12}\right)$

$= \cos\left(\frac{\pi}{12} + \frac{5\pi}{12}\right) = \cos\left(\frac{6\pi}{12}\right)$

$= \cos\left(\frac{\pi}{2}\right) = 0$

(c) $\sin(735°) = \sin(15°) = \sin(45° - 30°)$

$735 - 360 = 375$

$375 - 360 = 15$

$= \sin(45°)\cos(30°) - \cos(45°)\sin(30°)$

$= \frac{\sqrt{2}}{2}\left(\frac{\sqrt{3}}{2}\right) - \frac{\sqrt{2}}{2}\left(\frac{1}{2}\right)$

$= \frac{\sqrt{6}}{4} - \frac{\sqrt{2}}{4} = \frac{\sqrt{6} - \sqrt{2}}{4}$

(d) $\cos\left(\frac{13\pi}{12}\right) = \cos\left(\frac{5\pi}{6} + \frac{\pi}{4}\right) = \cos\left(\frac{5\pi}{6}\right)\cos\left(\frac{\pi}{4}\right) - \sin\left(\frac{5\pi}{6}\right)\sin\left(\frac{\pi}{4}\right)$

$\frac{13\pi}{12} = \frac{12\pi}{12} + \frac{\pi}{12}$ ✗

$= \frac{11\pi}{12} + \frac{2\pi}{12}$ ✗

$= \frac{10\pi}{12} + \frac{3\pi}{12} = \frac{5\pi}{6} + \frac{\pi}{4}$ ✓

$= -\frac{\sqrt{3}}{2}\left(\frac{\sqrt{2}}{2}\right) - \frac{1}{2}\left(\frac{\sqrt{2}}{2}\right)$

$= -\frac{\sqrt{6}}{4} - \frac{\sqrt{2}}{4} = \frac{-\sqrt{6} - \sqrt{2}}{4} = -\frac{\sqrt{6} + \sqrt{2}}{4}$

EXAMPLE 4

Suppose that A and B are angles which belong to quadrant II with $\sin(A) = 4/5$ and $\cos(B) = -5/13$. Evaluate $\sin(A - B)$.

$\sin(A-B) = \sin(A)\cos(B) - \cos(A)\sin(B)$

$= \frac{4}{5}\left(-\frac{5}{13}\right) - \cos(A)\sin(B)$

$\cos A = \pm\sqrt{1-\sin^2 A}$

$= -\sqrt{1-(4/5)^2}$

$= -\sqrt{1-16/25}$

$= -\sqrt{9/25}$

$= -3/5$

$\sin(B) = \pm\sqrt{1-\cos^2 B}$

$= +\sqrt{1-(-5/13)^2}$

$= \sqrt{1-25/169}$

$= \sqrt{144/169}$

$= 12/13$

$\sin(A-B) = \frac{4}{5}\left(-\frac{5}{13}\right) - \left(-\frac{3}{5}\right)\left(\frac{12}{13}\right) = -\frac{20}{65} + \frac{36}{65} = \frac{16}{65}$

Verifying Identities

The new sum/difference identities can sometimes be applied to confirm identities introduced earlier in this workbook.

EXAMPLE 5

If possible, use the angle sum/difference identities to confirm the following cofunction identities.

$\cos(90° - \theta°) = \sin(\theta)$

$\cos(90°-\theta°)$

$= \cos(90°)\cos(\theta°) + \sin(90°)\sin(\theta°)$

$= 0 \cdot \cos(\theta°) + 1 \cdot \sin(\theta°)$

$= \sin(\theta°)$ ✓

$\tan\left(\frac{\pi}{2} - \theta\right) = \cot(\theta)$

Cannot apply angle sum/difference identities here

$\tan(\pi/2 - \theta)$

$= \frac{\tan(\pi/2) - \tan(\theta)}{1 + \tan(\pi/2)\tan(\theta)}$

undefined!

In the next lesson, we'll be working with the double-angle identities. These are direct results of the angle sum identity.

EXAMPLE 6

Use the angle sum/difference identities to confirm the following double angle identities.

$\sin(2\theta) = 2\sin(\theta)\cos(\theta)$

$\sin(2\theta) = \sin(\theta+\theta)$

$= \sin(\theta)\cos(\theta) + \cos(\theta)\sin(\theta)$

$= 2\sin(\theta)\cos(\theta)$

$\cos(2\theta) = \cos^2(\theta) - \sin^2(\theta)$

$\cos(2\theta) = \cos(\theta+\theta)$

$= \cos(\theta)\cos(\theta) - \sin(\theta)\sin(\theta)$

$= \cos^2(\theta) - \sin^2(\theta)$

Careful... this is NOT the Pythagorean identity

Simplifying Expressions

Many times, complex expressions involving an angle sum or difference can be simplified significantly by applying an appropriate identity.

EXAMPLE 7

Simplify the expression.

$$\frac{\sec(x+\pi)}{\csc\left(\frac{\pi}{2}-x\right)} = \frac{\sin\left(\pi/2 - x\right)}{\cos(x+\pi)}$$

$$= \frac{\sin(\pi/2)\cos(x) - \cos(\pi/2)\sin(x)}{\cos(x)\cos(\pi) - \sin(x)\sin(\pi)}$$

$$= \frac{1\cdot\cos(x) - 0\cdot\sin(x)}{\cos(x)\cdot -1 - \sin(x)\cdot 0}$$

$$= \frac{\cos(x)}{-\cos(x)}$$

$$= -1$$